

Accuracy. Mostly accuracy specifications given in the manual are related to vertical DC accuracy. For example, DC accuracy of $\pm 3\%$ at 5mV/div is common among budget-friendly oscilloscopes. Due to various design and manufacturing challenges, it is generally not practical for a manufacturer to guarantee AC accuracy better than $\pm 3\text{dB}$.

Probe (active or passive). You can call it test lead or test prod or, simply, a probe. We often forget or overlook the importance of the probe while using an oscilloscope. You should select an oscilloscope probe that faithfully transmits the signal from your device under test to the oscilloscope.

Broadly, probes are of two types: active and passive. An active scope probe uses a high-impedance, high-frequency amplifier mounted in the probe head and a screened lead. A passive scope probe contains no active electronic part and therefore requires no external power. A typical passive probe uses a mega-ohm series resistor shunted by a low-value capacitor to make an RC-compensated divider with cable capacitance and scope input. As a rule of thumb, for a general-purpose oscilloscope with less than 500MHz measurements, a high-impedance passive probe is a good choice.

Attenuation refers to the output signal's ratio to the input signal with respect to voltage. Usual oscilloscope probe attenuation levels are 10 and 1 (1 for no attenuation). The 10x scope probe uses a 9-mega-ohm series resistor to provide 10:1 attenuation.

Maximum input voltage. Every electronic device has its limits when it comes to high voltage. If your signal exceeds maximum rated voltage, there is a good chance the scope will get damaged. Most general-purpose oscilloscopes have maximum AC voltage ratings up to 400V and DC ratings up to 300V.

Waveform update rate. It is an

Specifications	BitScope BS05U	Picotech 2204A-D2	Hantek 6022BE	Owon VDS1022
Bandwidth	20MHz	10MHz	20MHz	25MHz
Sampling rate	40MSa/s	100MSa/s	48MSa/s	100MSa/s
Number of channels	2	2	2	2+1 (multi)
Accuracy	$\pm 4\%$	$\pm 3\%$	$\pm 3\%$	$\pm 3\%$
Multifunction	Yes	Yes	Yes	Yes
Price	₹ 11,098	₹ 15,896	₹ 6998	₹ 10,380

Price as given on element14.com and ebay.in

important specification in oscilloscopes, which is expressed as waveforms per second (wfms/s). Higher waveform update rates provide better visual insight into signal behaviour.

Oscilloscopes have an inherent dead-time between each waveform acquisition. During the dead-time, any signal activity that may be occurring will be missed. Capturing random and infrequent events becomes a matter of probability, like rolling the dice. The higher the probability of capturing, the better the scope will display the results. Oscilloscopes with 50,000wfms/s or higher are easily available in the market.

External trigger. This is not used often but is sometimes very useful. The trigger controls let you stabilise repeating waveforms and capture single-shot waveforms. The trigger makes repeating waveforms appear static on the oscilloscope display.

Display size. Display size always matters as we see signal waveforms on the oscilloscope's screen. A bigger screen display is always more comfortable to watch and work with. There are traditional cathode ray tubes, thin-film-transistor liquid-crystal displays with touchscreen display, flexible display and so on.

Form factor. Oscilloscopes may be divided into three categories depending on size and shape: benchtop, handheld and PC based.

A benchtop oscilloscope is aimed at having the highest performance. Almost all functionalities are available in this category. Handheld is

portable and hence advantageous for the user on the move, but display quality and performance may be poor as compared to benchtop.

A PC based oscilloscope is becoming popular because PCs and laptops are easily available at home or workplace. You just need a portable external hardware or a dedicated add-on card in your PC along with suitable software.

Wireless supported. Wireless oscilloscopes support Bluetooth and Wi-Fi interface. The oscilloscope uses Wi-Fi module to create its own WLAN hotspot. Measurements can be done using the Web browser on a notebook, tablet or mobile phone.

With wireless, the PC becomes a virtual oscilloscope and can be used in special situations such as monitoring the signal while on the move or life-risky environment like in nuclear stations. But most wireless oscilloscopes have limited bandwidth.

IoT supported. Scopes having built-in wireless LAN module or Wi-Fi module act as hotspots. Some oscilloscopes have dedicated Ethernet ports for connecting to the Internet. Measurements can be monitored remotely using the Web browser on tablets, smartphones or PCs. The Internet of Things (IoT) capability on the scope allows measurements on high-voltage environments and brings safety to users.

Industrial automation, education management, remote test and measurement and some other functions need IoT oscilloscopes.

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